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Neural Networks and Deep Learning_CSCI_6366_80
Deliverable 01 - Project/Team Selection

Team for the project:

- **Pradnya Tendolkar**
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Problem statement:

Testing different deep learning models for electricity consumption forecasting. The task is to build an hourly electricity consumption predictor from the Datetime and PJME_MW (which is Energy Consumption in Megawatt) series values so that operators could schedule generation, plan reserves and price energy more reliably. The series exhibits strong daily, weekly, seasonal patterns, holiday effects and occasional shocks, so the challenge is to learn recurring patterns while staying resilient to non-stationarity, holiday effects and occasional shocks.

About the data:

I would be using the **PJM Hourly Energy Consumption** dataset from Kaggle which is originally from PJM Interconnection. It is a clean two columned time series dataset with an hourly **Datetime** index and **load in megawatts (MW)**. The sample spans approximately from **2001-12-31 to 2018-08-02** with typical demand ranging from **14.5k to 62k MW**. The figure below is a preview of PJME_hourly.csv

PJME_hourly.csv (4.07 MB)

Detail	Compact	Column
Datetime Date 2001-12-31 2018-08-02	# PJME_MW Megawatt Energy Consumption 14.5k 62k	2002-12-31 01:00:00 26498.0 2002-12-31 02:00:00 25147.0 2002-12-31 03:00:00 24574.0 2002-12-31 04:00:00 24393.0 2002-12-31 05:00:00 24860.0 2002-12-31 06:00:00 25000.0

Dataset Link:

https://www.kaggle.com/datasets/robikscube/hourly-energy-consumption?select=PJME_hourly.csv

Goal:

This project comes under time-series forecasting. Time series forecasting is a process of using historical data to predict future values in a sequence. As we can see PJM load shows strong autocorrelation. It shows clear daily, weekly seasonality and slower long-term trends. So sequence-aware models would be the right fit. I reviewed the papers listed below and identified several promising improvement strategies. I'll implement a few of the strongest methods on this dataset, moreover I will try to implement some new methods, and will train deep-learning models from scratch (e.g., LSTM, Transformer, and like others listed below in the papers) and evaluate them with standard metrics like MAE, RMSE and MAPE. The goal is to compare approaches fairly and select the best-performing deep learning architecture for this data.

Research for the project:

I have studied several related research papers to gather insights and I've listed them below along with the key takeaways

1) Research on building energy consumption prediction algorithm based on customized deep learning model

<https://energyinformatics.springeropen.com/articles/10.1186/s42162-025-00483-y>

Goal: Designed a building-energy forecaster that couples a customized CNN with Q-Learning (CCNN-QL) and show it outperforms strong baselines. Additionally, demonstrated on public benchmarks (such as UCI IHEPC and PJM) and discussed deployment implications.

Dataset: Their own collected dataset from non-residential buildings across six Chinese cities (Handan, Tianjin, Dalian, Beijing, Changzhou, Shanghai) with 12 attributes incl. indoor and outdoor conditions and energy use in kWh. Outliers handled via three-sigma, means for missing values and min-max scaling

Benchmarks: UCI IHEPC (household power, France, 2006–2010) and **PJM** hourly regional loads (MW).

Methods & Models used:

- **Proposed model:** CCNN-QL, a CNN with five conv + five max-pool layers (ReLU) integrated with model-free Q-Learning for decision-aided forecasting.
- **Training and eval:** 80/20 split, 7-fold CV, metrics: MSE, RMSE, MAE.
- **Baselines:** CNN, CNN-LSTM, CNN-BiLSTM, CNN-GRU

Results (high level)

- On author's dataset: CCNN-QL achieved MSE 0.011, RMSE 0.121, MAE 0.097, beating all baselines.
- On IHEPC (hourly): best with MSE 0.0002, RMSE 0.0141, MAE 0.0033; daily: MSE 0.0005, RMSE 0.0223, MAE 0.0045.
- On PJM (examples): AEP MSE $8e-7$, RMSE 0.0008942, MAE 0.0000104;
DAYTON MSE $1.1e-6$, RMSE 0.0010488, MAE 0.0001078;
COMED MSE $1.2e-6$, RMSE 0.0010954, MAE 0.0000592 (best across compared models)

Best performer: CCNN-QL, consistently top on the in-house set, IHEPC (hourly and daily) and multiple PJM zones

Value: The CCNN-QL integration extracts rich features and supports adaptive, real-time energy management. Authors note clear wins in accuracy but flag computational cost and scalability of deep CNNs + Q-Learning

2) Deep learning-based forecasting of electricity consumption

<https://www.nature.com/articles/s41598-024-56602-4>

Goal: Developed a deep-learning time-series model for forecasting building electricity consumption and optimize it via different training optimizers/hyperparameters to improve accuracy.

Dataset: Hourly electricity-use data for one year (8,760 hours) from a hospital building in Phoenix, USA, sourced from OpenEI. Target is kWh at hourly resolution. Exploratory analysis shows weekday and weekend patterns, daily and weekly trends.

Methods & Models:

Model: LSTM regression network implemented in TensorFlow/Keras. Inputs are sliding windows of 60 past hours, data is normalized and shaped as (samples, time-steps, features)

Train and test: 90% train and 10% test with an internal validation split to curb overfitting.

Objective: maximize R^2 and minimize MAE

Optimization: Comparing multiple optimizers such as SOG, Nadam, Adadelta, FTRL, RMSprop, Adam. Moreover, tuning of layers, neurons, activation, dropout and epochs.

Results: Progressive tuning improves performance across six "settings". Reported R^2 rises from -5% to 95%

Best result: $R^2 = 0.95$ under the strongest (sixth) setting. The paper's optimizer table maps this setting to Adam

Value: LSTM can accurately forecast short-term building electricity consumption on real data. Optimizer choice and hyperparameter tuning materially improve accuracy. Authors suggest scaling to larger, multi-organization datasets as future work

3) Energy consumption prediction using modified deep CNN-Bi LSTM with attention mechanism

<https://www.sciencedirect.com/science/article/pii/S240584402417538X>

Goal: Proposed a modified Deep CNN–BiLSTM with an attention mechanism to improve short-term household energy consumption forecasting (hourly, daily, monthly) and compare it against strong baselines while also reporting training prediction time.

Dataset used: Household Energy Consumption (UCI/IHEPC) time-series (also referenced via Kaggle): multi-year residential electricity measurements (e.g., voltage, global intensity, sub-meterings). The authors resample to hourly/daily/monthly horizons and describes a 4-year sampling window.

Methods & models

- **Preprocessing:** handle missing values and feature scaling, created hourly/daily/monthly resamples for supervised regression.
- **Proposed model:** Deep CNN + BiLSTM + Attention (dropout, max-pool, dense). CNN extracts “spatial” patterns. BiLSTM captures temporal dependencies. Attention weights time steps/features most relevant to the forecast.
- **Training and eval:** 80/20 split. Metrics used were MSE, RMSE, MAE, MAPE

Baselines compared: LSTM, GRU, Bi-LSTM, Attention-LSTM, CNN-LSTM. Additional comparisons vs. ESN, EECP-CBL and classic models (KNN, SVM, ANN).

Results: On the UCI household series, the proposed CNN-BiLSTM+Attention attains lower errors than all listed baselines across horizons. MSE = 0.1234, RMSE = 0.022, MAE = 0.1234, MAPE = 32.42. Comparative tables also show the proposed model outperforming LSTM, GRU, Bi-LSTM, Attention-LSTM, CNN-LSTM and others (e.g., EECP-CBL, ESN, KNN, SVM, ANN) on RMSE, MSE, MAE and having competitive training/inference time.

Best model: The modified Deep CNN-BiLSTM with attention is the top performer among all methods evaluated by the authors.

Value: The hybrid CNN-BiLSTM+Attention improves short-term residential load forecasting with lower error metrics and reasonable compute time.

4) Predicting Power Consumption Using Deep Learning with Stationary Wavelet

Goal: Evaluated whether stationary wavelet transform (SWT) denoising, especially the bior2.4 mother wavelet, combined with deep learning can improve short-term household power-consumption forecasting across multiple time resolutions and identify the best wavelet–model pairing.

Dataset: UCI IHEPC (Individual Household Electric Power Consumption): minute-level residential electricity data (2006–2010). Following a supervised sliding-window setup that uses the previous 15 time steps to predict the next value, the authors evaluate models at 1-minute, 15-minute, and 1-hour sampling intervals, using an 80/20 train–test split

Methods & Models:

- **Preprocess:** cleaned anomalies, missing, duplicates and used MinMaxScaler normalization.
- **Wavelet denoising (SWT):** compared bior2.4, rbio2.4, coif2, db2, sym2. Kept same length as original series and then reconstruct.
- **Deep learning models:** LSTM, GRU, Bi-LSTM, Bi-GRU, Bi-GRU-LSTM (Keras and TensorFlow up to 15 epochs) trained on raw series and on SWT-denoised series to quantify gains.

Results:

Without SWT (raw series):

- 1-min: LSTM RMSE 0.21, MAE 0.07, MAPE 8.55.
- 15-min: errors around RMSE 0.46–0.47, MAE 0.28–0.29, MAPE 37.8–43.6.
- 1-hour: errors around RMSE 0.51–0.52, MAE 0.36–0.38, MAPE 47–55.

With SWT denoising (bior2.4 best in most cases)

- 1-min: bior2.4/GRU: RMSE 0.06, MAE 0.03, MAPE 5.65;
bior2.4/Bi-GRU-LSTM: 0.07, 0.03, 5.09.
Gains vs. raw DL often >60% on RMSE/MAE.
- 15-min: bior2.4/GRU: 0.15, 0.10, 13.62;
bior2.4/LSTM: 0.15 / 0.10 / 13.55.
66% error reductions vs. raw DL.
- 1-hour: bior2.4/Bi-GRU: 0.17, 0.13, 17.53
bior2.4/GRU: 0.18, 0.13, 18.21
Authors note GRU is faster, so bior2.4/GRU is the most efficient choice.

Best Result : Across time scales, SWT(bior2.4) + GRU delivers the most consistent accuracy.

Value: Wavelet denoising (especially bior2.4 SWT) materially boosts deep models for load forecasting. Capturing both short-term fluctuations and long-term trends and improvements persist from 1-min to 1-hour horizons (with shorter intervals being easier). Denoising helps the nets focus on meaningful patterns and reduces error by 60–70% relative to raw deep-learning baselines.

5) State of the art in energy consumption using deep learning models

<https://pubs.aip.org/aip/adv/article/14/6/065306/3295871/State-of-the-art-in-energy-consumption-using-deep>

Goal: Forecasting monthly national electricity consumption for Brazil, Canada and France using deep-learning sequence models. Compared LSTM variants vs. GRU and then used the best model per country to predict all months of 2017.

Dataset used: Monthly GWh consumption from the IEA Monthly Electricity Statistics, covering 2006–2016. Train/test split per country: 100 months for training (Jan 2007–Apr 2015) and 20 months for testing (May 2015–Dec 2016)

Methods & models:

- **Models compared:** LSTM, Stacked LSTM, Bidirectional LSTM (BiLSTM), GRU
- Supervised setup on the monthly series
- Metrics on test set: R^2 , MSE, RMSE, MAE, MAPE and best architecture was selected per country
- It was note that with relatively few data points, simpler models (fewer parameters) can generalize better. Model choice may also relate to generation-mix heterogeneity.

Results:

- Brazil: LSTM: R^2 0.54, MSE 531,302, RMSE 728.90, MAE 610.01, MAPE 0.02.
- Canada: GRU : R^2 0.86, MSE 4,328,169, RMSE 2,080.43, MAE 1,688.29, MAPE 0.04.
- France: GRU: R^2 0.84, MSE 7,181,056.85, RMSE 2,679.75, MAE 2,187.29, MAPE 0.05.

Value: Deep learning models are suitable for national-level energy forecasting with limited monthly data. GRU performed best for Canada & France while LSTM fit Brazil better, potentially due to Brazil's more heterogeneous and renewable-driven generation profile.

6) Optimized deep neural network architectures for energy consumption and PV production forecasting

<https://www.sciencedirect.com/science/article/pii/S2211467X25000677>

Goal: Proposed a hybrid GA-PSO approach to optimize Deep Neural Network (DNN) architectures for energy-load/PV forecasting and showed it beats common hyperparameter optimizers (Bayesian Optimization and Evolutionary Strategy) and standard ML baselines.

Dataset used: Twelve PJM hourly electricity-consumption datasets (AEP, COMED, DAYTON, DEOK, DOM, DUQ, EKPC, FE, NI, PJM, PJME, PJMW) which were drawn from PJM's website. The paper describes the data as decade-scale hourly series. The table has two columns: Datetime and PJME-MW.

Methods & models:

- ODNN: a DNN whose architecture and hyperparameters are jointly tuned by Genetic Algorithm + Particle Swarm Optimization (GA-PSO). GA explores globally and PSO refines locally
- Compared ODNN to an untuned Basic DNN (BDNN) and to XGBoost, Random Forest, k-NN, SVM, Decision Tree, Linear Regression. Also compared GA-PSO against Bayesian Optimization and Evolutionary Strategy on optimization benchmarks and DNN tuning.

Results:

- Across PJM zones, ODNN consistently lowers error versus BDNN and the ML baselines on RMSE, MAE, R^2 , MedAE, EV, RAE.
- Relative Absolute Error (RAE) likewise favors ODNN across most zones.
- The paper summarizes the overall improvement as +27% accuracy / –22% error across metrics and tasks.

Value: Hybrid GA-PSO effectively balances exploration and exploitation to discover stronger DNN architectures for energy forecasting, improving both accuracy and efficiency. The authors note limitations: computational overhead, lack of real-time integration and dataset-dependence. They suggest parallelization, pruning, quantization and transfer learning for future work.

7) Intelligent deep learning techniques for energy consumption forecasting in smart buildings: a review

<https://link.springer.com/article/10.1007/s10462-023-10660-8>

Goal: Provided a comprehensive review of deep-learning methods for forecasting building energy consumption. Analyzed datasets, loads, metrics and surface challenges and research gaps. The authors also sketch a multivariate hybrid approach they suggest can improve accuracy.

Results:

- TCNs are reported to outperform LSTMs in some studies.
- Handling peaks benefits from hybrids such as EMD+ELM and wavelet-based preprocessing, collaborative representations can help.
- Overall, DL and hybrid DL methods generally beat traditional ML for complex, high-resolution building loads. No single model is universally “best”. Performance depends on dataset, horizon and features.

Value: Multivariate and hybrid architectures shows strong promise for improving building load forecasts but standardized datasets, benchmarks, better peak-load modeling and clearer guidance on method selection remain open needs.

8) Deep Learning with Multi-Headed Attention for Forecasting Residential Energy Consumption

<https://ijisae.org/index.php/IJISAE/article/view/3563>

Goal: Proposed a CNN–LSTM forecaster enhanced with multi-headed attention to capture short- and long-term dependencies in household energy use and showed it improves accuracy, especially around transient and impulsive peaks

Dataset used: UCI Individual Household Electric Power Consumption (IHEPC): minute-level residential data. Authors also analyze multiple temporal resolutions (1, 15, 30, 45 minutes and 1 hour).

Methods & models:

- **Architecture:** 1D-CNN (local/spatial patterns), LSTM (temporal memory) and multi-headed attention (selectively weights salient timesteps and features). Direct multi-step forecasting avoids recursive error accumulation. Min-max normalization, sliding windows, MSE loss, gradient-descent training.

Baselines: Classical LR, ARIMA, DT, RF, SVR, and neural CNN, LSTM, CNN-LSTM. Attention variants (none, single-head, multi-head) on CNN, CNN-LSTM, LSTM. 10-fold CV for error reporting

Results:

- **Across time resolutions (MSE):** Proposed model generally beats classical ML and untuned DL.
- **Attention ablation:** On CNN-LSTM, multi-headed attention cuts MSE to 0.0416 vs 0.0560 (single-head) and 0.0576 (no attention) evidencing the gain from multi-head weighting
- So the proposed approach reduces error by 22.82% vs a conventional CNN-LSTM baseline.

Best result: CNN–LSTM with multi-headed attention is the top performer among tested variants and attention settings and outperforms classical ML across evaluated horizons.

Value: Combining CNN + LSTM with multi-headed attention yields more accurate and robust residential load forecasts, capturing both local patterns and long-range temporal structure, improving peak handling and mitigating multicollinearity effects. The authors suggest extending to additional contextual data and exploring alternative attention designs.